

Experiment 4

AIM: Use npm to install and manage packages

Step 1: Download from <https://git-scm.com>

Step 2: C:\Users\Admin\npm-lab-1> \$ node -v

v22.13.1

C:\Users\Admin\npm-lab-1> -v

10.9.2

C:\Users\Admin\npm-lab-1> \$ mkdir npm-lab-

1 C:\Users\Admin\npm-lab-1> \$ cd npm-lab

C:\Users\Admin\npm-lab-1> \$ npm init -y

Wrote to c:\npm-lab\package.json:

```
{
  "name": "npm-lab-1",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" fifi exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

Step 3: Install Required Package



```
C:\Users\Admin\npm-lab-1> $npm install express
```

```
added 65 packages, and audited 66 packages in 57s
```

```
22 packages are looking for funding
```

```
run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
C:\Users\Admin\npm-lab-1>$ npm install axios
```

```
added 11 packages, and audited 77 packages in 3s
```

```
24 packages are looking for funding
```

```
run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
C:\Users\Admin\npm-lab-1>$ npm install nodemon --save-dev
```

```
added 26 packages, and audited 103 packages in 4s
```

```
29 packages are looking for funding
```

```
run `npm fund` for details
```

```
found 0 vulnerabilities
```

Step 4:Create Web Server

Create a file named index.js and Add below content :

```
const express = require("express");
```

```
const axios = require("axios");
```

```
const app = express();
```

```
app.get("/", (req, res) =>
{ res.send("Hello Open Source lab");
});
```

// JSON Route

```
app.get("/api/info", (req, res) =>
{ res.json({
name: "NPM lab",
subject: "Open Source Technology",
semester: "BE 6th Sem"
});
});
```

// Axios Demo Route

```
app.get("/api/user", async (req, res) => {
const response = await axios.get("https://jsonplaceholder.typicode.com/users/1");
res.json(response.data);
});
```

```
app.listen(3000, () => {
console.log("Server running at http://localhost:3000");
});
```

Step 5: Run Server (Normal Way)

node index.js

Open browser:

http://localhost:3000

`http://localhost:3000/api/info`

`http://localhost:3000/api/user`

Step 6: Add npm Scripts

Open package.json and modify scripts section:

```
"scripts": {  
  "start": "node index.js",  
  "dev": "nodemon index.js"  
}
```

Step 7: Run Using npm Script

1. Normal mode:

`npm run start`

2. Development mode (nodemon):

`npm run dev`

Now modify index.js text and save → Server automatically restarts (nodemon demo).

Step 8: Delete node_modules and Reinstall

Delete manually

Now **reinstall**:

`npm install`

node_modules folder recreated

Dependencies installed from package.json

Step 9: Add .gitignore File

PS D:\npm-lab-1> notepad .gitignore (open notepad write this after that save and close)

node_modules/



Step 10: Upload Project to Git (Without node_modules)

```
C:\Users\Admin\npm-lab-1>$ git init
```

Initialized empty Git repository in D:/npm-lab-1/.git/

```
C:\Users\Admin\npm-lab-1> (master)$ git config --global user.name "hemanshijalu126463-prog"
```

```
C:\Users\Admin\npm-lab-1> (master)$ git config --global  
user.email "hemanshi.jalu126463@marwadiuniversity.ac.in"
```

```
C:\Users\Admin\npm-lab-1> (master)$ git add . git
```

```
commit -m "Initial commit"
```

```
git branch -M main
```

```
git push -u origin main
```

From <https://github.com/hemanshijalu126463-prog/OST>

```
* branch          main      -> FETCH_HEAD
```

Already up to date.

Enumerating objects: 9, done.

Counting objects: 100% (9/9), done.

Delta compression using up to 8 threads

Compressing objects: 100% (7/7), done.

Writing objects: 100% (8/8), 12.50 KiB | 12.50 MiB/s, done.

Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)

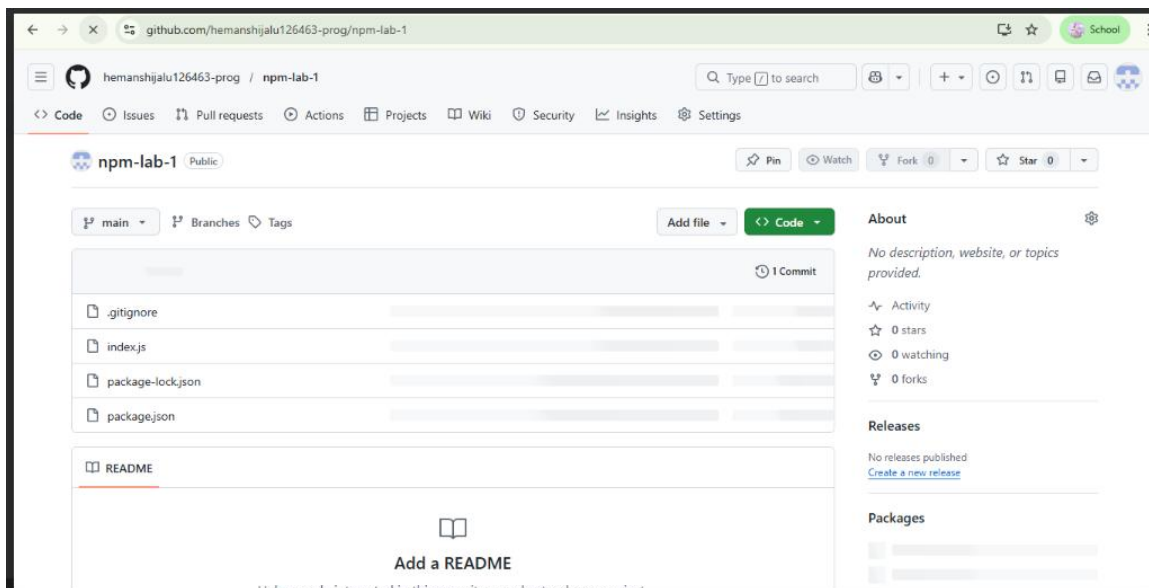
remote: Resolving deltas: 100% (3/3), completed with 1 local object.

To <https://github.com/hemanshijalu126463-prog/OST.git>

```
749ef93..9ca76e8 main -> main
```

branch 'main' set up to track 'origin/main'.

Output:



```
Command Prompt

C:\Users\Admin>mkdir npm-lab-1

C:\Users\Admin>cd npm-lab-1

C:\Users\Admin\npm-lab-1>npm init -y
Unknown command: "init-y"

Did you mean this?
  npm init # Create a package.json file
To see a list of supported npm commands, run:
  npm help

C:\Users\Admin\npm-lab-1>npm init -y
wrote to C:\Users\Admin\npm-lab-1\package.json:

{
  "name": "npm-lab-1",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\Admin\npm-lab-1>npm install express
added 65 packages, and audited 66 packages in 1m

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Admin\npm-lab-1>npm install axios
added 11 packages, and audited 77 packages in 3s

34 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  + v

PS C:\Users\Admin\npm-lab-1> git init

Reinitialized existing Git repository
Reinitialized existing Git repository in C:/Users/Admin/npm-lab-1/.git/
PS C:\Users\Admin\npm-lab-1> git add .
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF t
warning: in the working copy of 'package.json', LF will be replaced by CRLF the ne
[main (root-commit) c2ab223] Initial commit
 4 files changed, 989 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index.js
 create mode 100644 package-lock.json
 create mode 100644 package.json
PS C:\Users\Admin\npm-lab-1> git branch -M main
PS C:\Users\Admin\npm-lab-1> git remote add origin https://github.com/hemanshijalu12
6463-prog/npm-lab-1
PS C:\Users\Admin\npm-lab-1> git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 9.11 KiB | 1.82 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hemanshijalu126463-prog/npm-lab-1
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Admin\npm-lab-1> 
```